Module 5 Lecture - Sensation and Perception

Introductory Psychology

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1 Overview and Introduction

1.1 Textbook Learning Objectives

- Distinguish between sensation and perception
- Describe the concepts of absolute threshold and difference threshold
- Discuss the roles attention, motivation, and sensory adaptation play in perception
- Describe important physical features of wave forms
- Show how physical properties of light waves are associated with perceptual experience
- Show how physical properties of sound waves are associated with perceptual experience
- Describe the basic anatomy of the visual system
- Discuss how rods and cones contribute to different aspects of vision
- Describe how monocular and binocular cues are used in the perception of depth
- Describe the basic anatomy and function of the auditory system
- Explain how we encode and perceive pitch
- Discuss how we localize sound
- Describe the basic functions of the chemical senses
- Explain the basic functions of the somatosensory, nociceptive, and thermoceptive sensory systems
- Describe the basic functions of the vestibular, proprioceptive, and kinesthetic sensory systems

1.2 Instructor Learning Objectives

- Be able to connect sensation and perception back to concepts in biopsychology, consciousness/sleep, and gestalt psychology
- Explain how the anatomical structure gives rise to functions of perception, much like we did on focusing with brain structures and their respective functions

1.3 Introduction

Discuss: Close your eyes for a moment and try to be quiet - think about what you hear, smell, feel, taste. Describe some of what you felt here

- What you just wrote down are ____________ that are being received by your senses (without sight)
- Those senses are taking in the information in our environment and about our body and, put together, give rise to our ____________ experience.

Discuss: Describe what it means to be conscious, based upon the discussion we had in Module 4

- Today is all about examining how these different senses work both ____________ and psychologically

2 Sensation versus Perception

2.1 Introduction

- Sensation and perception ____________ sound quite similar, but they have some important distinctions in psychological research

2.2 Sensation

- **Sensation** occurs via **sensory receptors**, which are a specialized type of ____________.
  - The sensory receptors recieve the actual stimuli, whether that is light, chemical, pressure, etc., and then pass that along to other neurons as an ____________ potential
  - Going from sensory stimuli in its ____________ form to an action potential is called **transduction**
- We have 5 basic senses:
  - ____________
  - ____________ (Audition)
  - ____________ (Olfaction)
  - ____________ (Gustation)
  - ____________ (Somatosensation)

? What sense is NOT routed through the thalamus

- (●) A) Smell
- ( ) B) Taste
- ( ) C) Touch
- ( ) D) Hearing

Explanation:

Discuss: Review: What lobe of the brain's cortex is most associated with hearing? Where is it located on the human head?

- We have several other slightly less talked about senses:
  - ________ sense (balance)
  - ________ and kinesthesia (body position and movement)
  - ________ (pain)
  - ________ (temperature)
- Our senses detect ________ in the environment, and whether this change is registered in our brain is determined by a sense's particular **Absolute threshold**
  - The absolute threshold is the amount of energy necessary for a sense to be triggered ________ percent of the time
  - Put another way, a stimulus must reach an absolute threshold for us to become ________ of it 50 percent of the time
  - Example: how ________ can a light be where my eyes will still detect light half of the time

Discuss: Rewrite that last example regarding the eye and light, but now applied to smell.

- Not all stimuli is ______ to rise to the level of consciousness, these stimuli are called **subliminal messages**.

Important

Subliminal messages are popularized as having an unconcious and pervasive impact on our behavior, but modern science has largely rejected that notion.

- Another way of understanding differences in stimulus ______ is through **just noticeable difference (jnd) or difference threshold**
  - This model looks at how the same stimulus may be noticeable or unnoticeable in different contexts
  - Example: On my hand, I can feel two pin pricks when they aren't particularly far apart, but cannot tell that same difference on my kneecap

? What part of the parietal lobe cortex is set up with different area amounts for sensory input for different parts of the body?

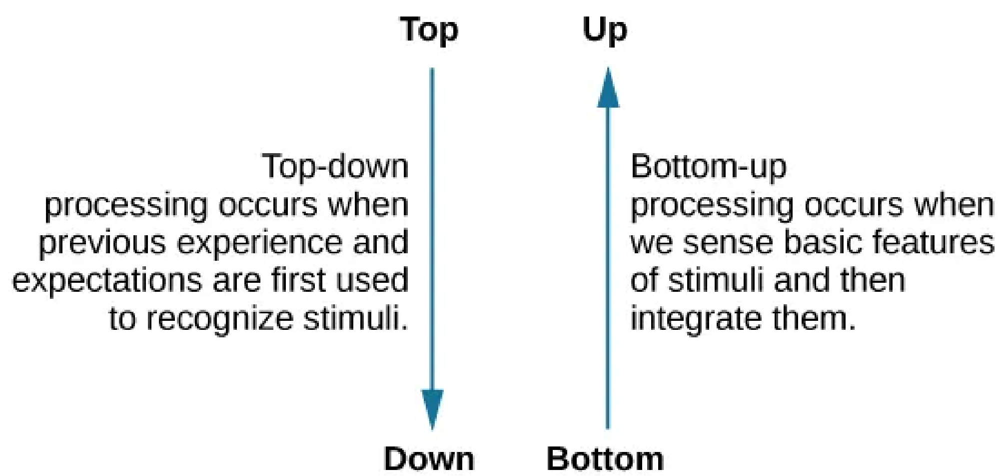
- (●) A) Broca's area
- ( ) B) Wernicke's area
- ( ) C) Somatosenroy cortex
- ( ) D) Longitudinal fissure

Explanation:

2.3 Perception

- Senses are the ______ inputs our body takes in and sends as an action potential signal, but **perception** is our actual complex experience and ______

of that information
  - Perception comes in two flavors: bottom-up and top-down
  - But, ________ are constantly active and equally important
- **Bottom-up processing** is taking sensory information and ________ it in our experience
  - It tends to be ________ and outside conscious control
  - Example: somebody drops a large metal water bottle on the floor in the quiet library, everyone reacts to look at it instantaneously out of curiosity or anxitey
- **Top-down processing** is the use of ________ and experiences to recognize stimuli
  - Example: your instructor recognizes subtle signs of possible aggression in a patient as I've been trained to closely observe



> Discuss: Try to come up with both an additional example of bottom-up and top-down processing

2.4 Sensations and Perceptions Together

- Recap:
  - Sensation can be though of the basic, biological, ________ process of taking in information
  - Perception is the psychological recognition and understanding of the senses

  - Thus, without sense ________, we cannot have perception
- There are many reasons our senses may ________ in different perceptions
  - Adaptation, attention, motivation (all discussed below)
  - Personality, culture, beliefs, expectations, experiences
- Senses, even when they may initially reach the absolute threshold, may ________ always rise to perception, due to **sensory adaptation**
  - An example of this would be when you first walk outside into bright light from a dark room. The blinding light may initially be very prominent to you, but, over time, your become accustomed to it
- Our **attention** is also critical to what is actually ________ in perception
  - Example: If you are on your phone while the instructor talks, you might ________ the noise, but you will not actually perceive it
  - In vision, missing something that is visually sensed, but not ________ is called **Inattentional blindness**

> **Important**
>
> Many people mistake attention problems for memory problems - but many people don't realize that they are simply not paying enough attention to perceive what they are missing!

- Our **motivation** also plays a similar role to ________: we are more likely to perceive something important to us
  - I am likely to notice my phone buzzing in my pocket, even in a busy room, because I am interested and excited by whatever is going off on my phone

> **?** Which of the following things is a new mother most likely to perceive, considering the contributions of motivation?
>
> - (●) A) Phone ringing
> - ( ) B) Baby crying
> - ( ) C) Sounds of conversation in a coffee shop
> - ( ) D) Smell of food
>
> Explanation:

Important

These things are very relevant to learning in classrooms! Your attention and motivation determine whether you will perceive anything at all!

3 Waves and Wavelengths

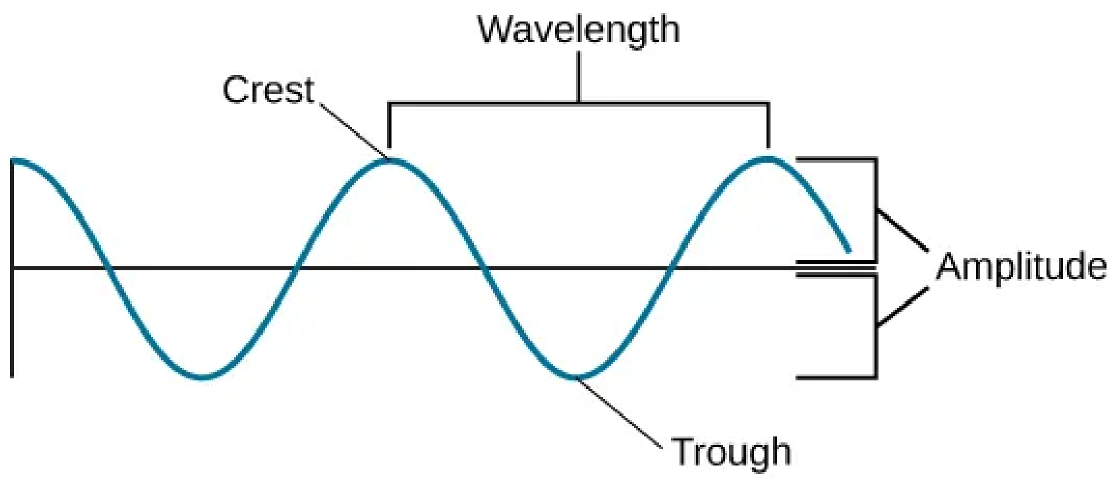
3.1 Introduction

- To understand visual and auditory senses, we have to understand ____________
  - That is because both light and audio are ____________ waves

Discuss: What other (recent) topic in class dealt with waves?

3.2 Basics of Amplitude and Wavelengths

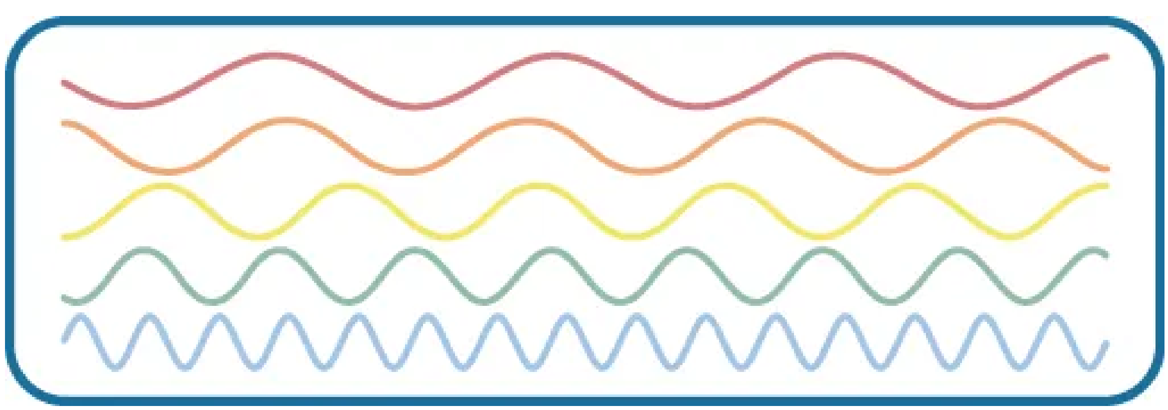
- **Amplitude** is the relative **crests (peaks)** and **troughs (dips)** of a wave
- **Wavelenth** is the length of one ____________ to the next
  - **Frequency**, usually measure in **hertz (Hz)**, is the number of waves that occur in a given time span
  - Thus, wavelength is ____________ correlated with frequency



? Which of the following colored waves appears to have the longest wavelength?

- (•) A) Orange
- ( ) B) Yellow
- ( ) C) Green
- ( ) D) Dark red

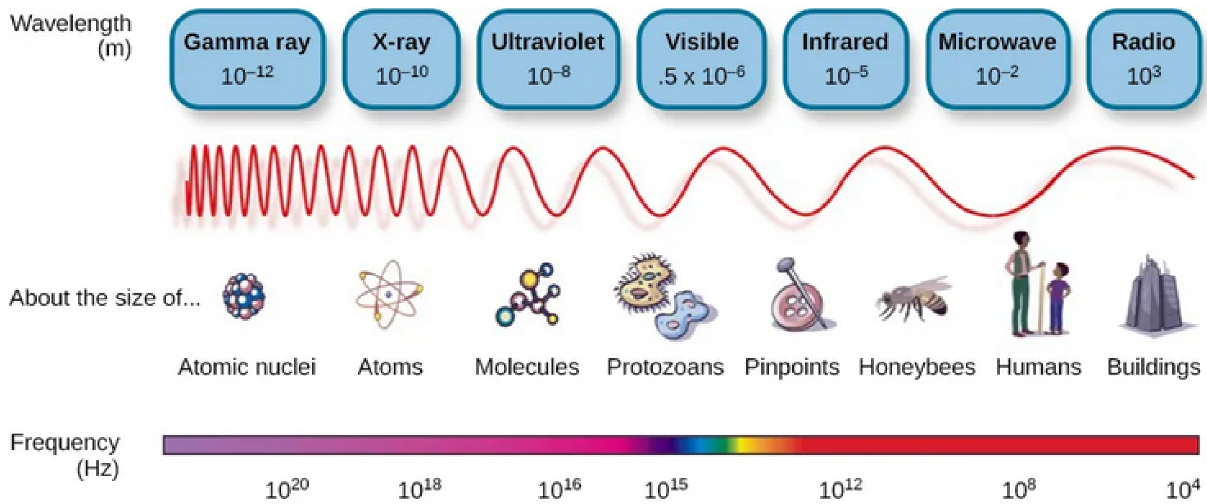
Explanation:



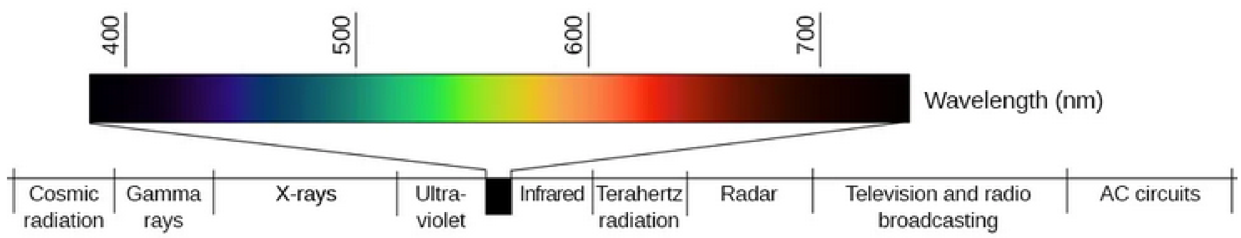
3.3 Light Waves

- We can't see ________________, but all of what we can see is called the **visible spectrum**

- The visible spectrum sits within a ________ **electromagnetic spectrum**, and is a relatively small portion, i.e., a lot more is invisible than visible



- Within that visible spectrum, we have our perception of ________ associated with different wavelengths



- Higher ________ of light waves correspond to brighter light / more intense color

3.4 Sound Waves

- With sound, we often identify things by **pitch**: which is how "low" or "high" it sounds
  - A ________ pitch sound is associated with low-frequency
  - A ________ pitch sound is associated with high-frequency

Discuss: Based on the definitions of frequency and wavelength given before, would we expect the wavelength of high-frequency sounds to be relatively short or long? Explain.

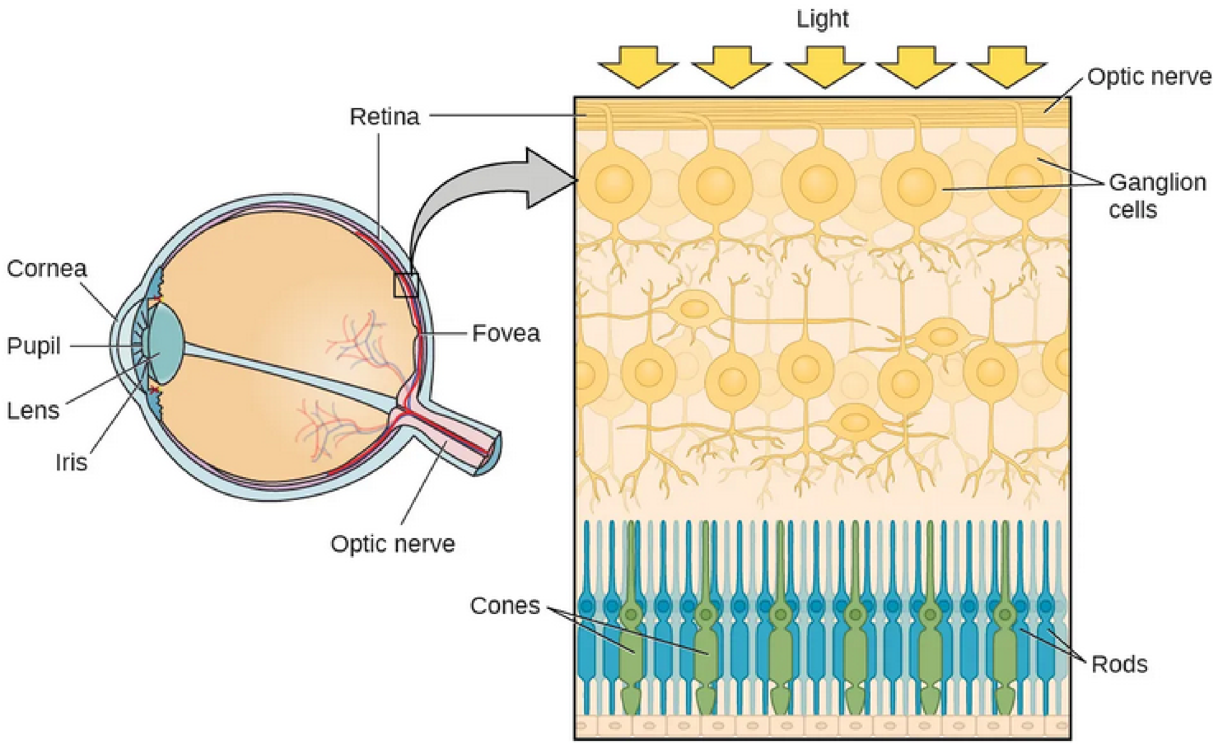
- Just like with the visible spectrum, we have an **audible spectrum**, outside which we cannot ________ certain sounds
- In the case of audio, a higher amplitude of wave ________ sound

4 Vision

4.1 Introduction

- For most individuals, vision is the ________ way we navigate and first interpret the world around us
  - Our specialized **eye** organ and associated structures allow us to ________, while our connected brain structures aid in perception

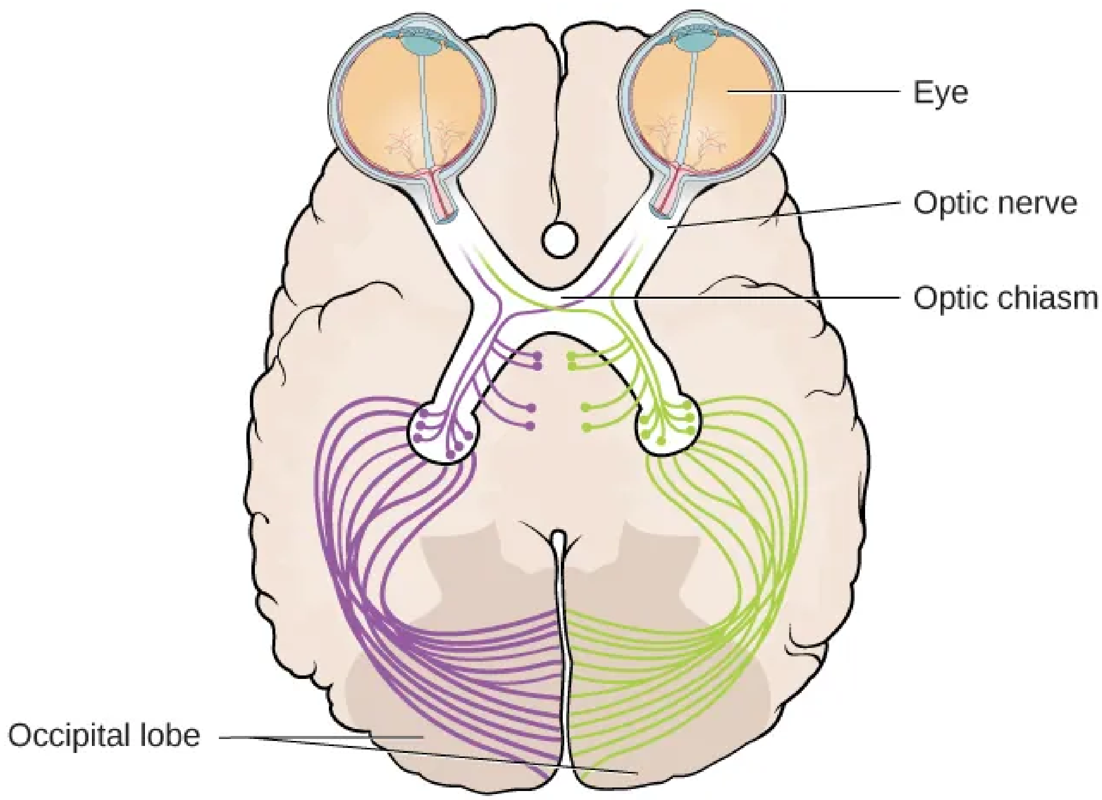
4.2 Anatomy of the Visual System



- The above image show the essential components of the human eye
- The structure and function of the eye parts are as follows:
  - **The cornea** is the outermost, transparent layer of the eye that both is a physical ______ barrier and helps focus light
  - The **pupil** is the small ______ (black hole) in our eye that can dilate or contract to let more or less light in
  - The pupil's size is controlled by ______ in the **iris**, where we have color.
  - The **lens** within the pupil does even more ______, hitting the **retina** on the back of the eye
  - The retina contains the **fovea**, contains specialized sensory ______ called **photoreceptor cells**
- Photoreceptor cells are one of two types:
  - **Cones** are mostly clustered in the ______ and help detect color, detail and dimension, but are especially light sensitive, i.e., don't work well in the dark
  - **Rods** are throughout the ______ and function better in low-light conditions to capture non-colored, non-detailed images

> **Important**
>
> This is why we can't see detail or color very well in the dark, because the cones aren't activated as much then, and we have to rely on our rods!

> **?** Review: The pupils normally dilates during periods of high arousal and alertness - what segment of the autonomic nervous system is active during times of high arousal?
>
> - (●) A) Parasympathetic
> - ( ) B) Central
> - ( ) C) Peripheral
> - ( ) D) Sympathetic
>
> Explanation:

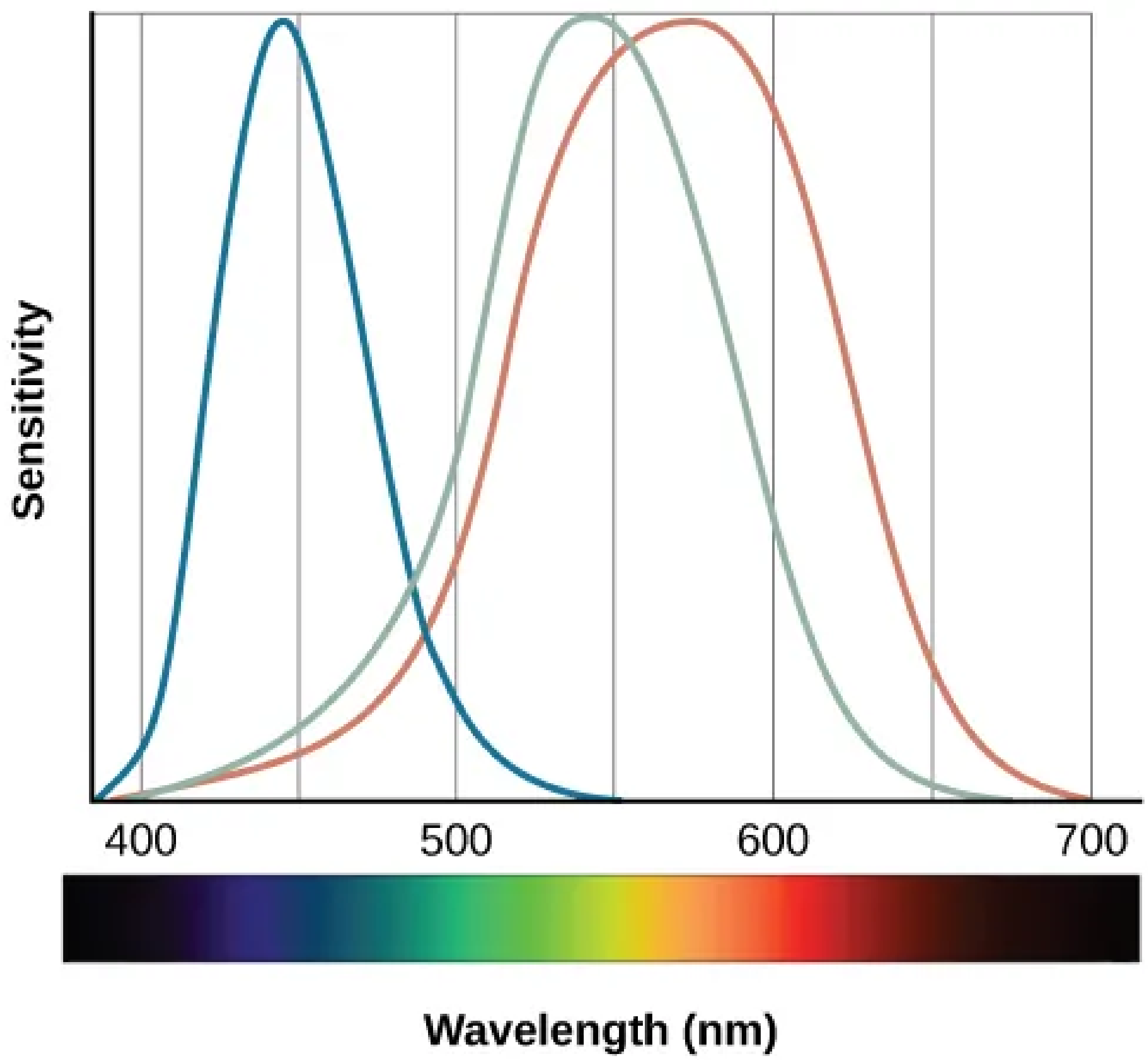


- Ganglion cells passed the signals from the ________ back through the **optic nerve**. The placement of this optic nerve creates a natural **blind spot** in our vision

  - BUT, we don't notice this because our two eyes have __________, non-overlapping blindspots and our brain works to not perceive them
- Pay careful attention to which side of each eye goes to which side of the occipital lobe

4.3 Color Perception

- The most prominent theory and explanation for color vision follows the **trichromatic theory**
  - This states that all visible colors are a __________ of red, green, and blue, resulting from 3 distinct cone types attuned to each of those colors
  - Evidence that this happens primarily on the __________



- Alternatively, there is **opponent-process theory** that suggest colors are perceived as one end of the pairs black-white, yellow-blue, and green-red.

- Evidence that this happens on the optic ______

Discuss: We call these theories - what does that mean and how does that differ from a hypothesis?

4.4 Depth Perception

- **Depth perception** allows us to see in 3 ______, and see distances properly in space

Important

Without any proper depth perception, the world would look as flat as a painting!

- We have two types of "clues" that allow for depth perception:
  - **Binocular cues** need ______ eyes
  - **Monocular cues** need only ______ eye
- In the case of ______ cues, we rely on things like **binocular disparity**, or the difference in sense between our eye
- For **monocular cues**, we can make sense of some ______ even in a two-dimensional representation by looking at how things overlap or show relative size.

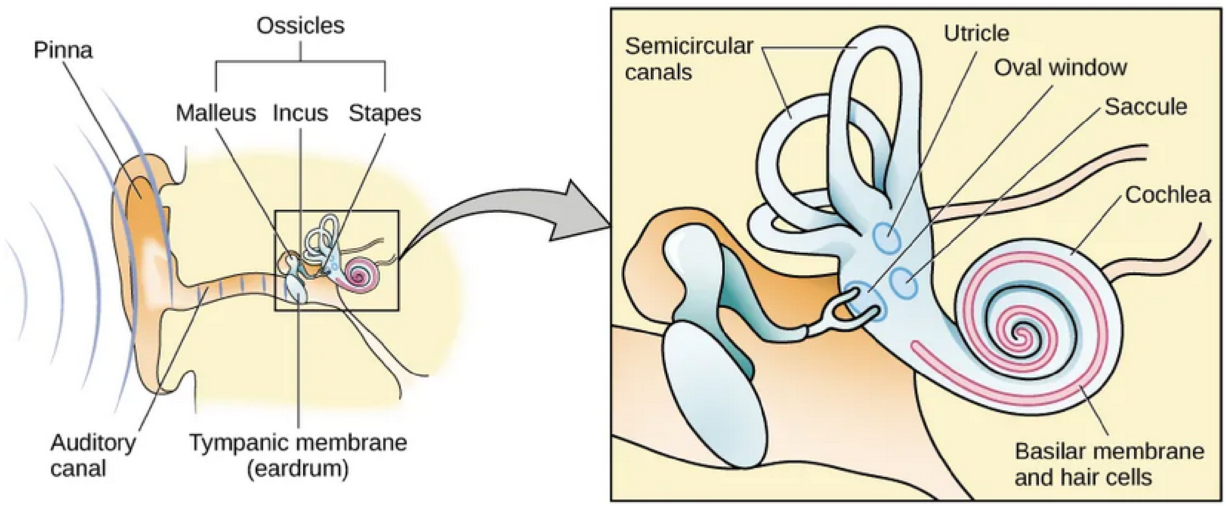
5 Hearing

5.1 Introduction

- Much like the sense of **sight** comes from reception of ______ waves, the sense of hearing comes from reception of ______ waves

- However, this process also involves air ________________, and a complex system of bones and membranes, just like the eye had several components

5.2 Anatomy of the Auditory System



- Our ear has several important parts:
  - **Pinna**, the ________________, protruding, wiggly part of our ear that focuses sound into..
  - The **auditory canal** which ________________ sound to the **tympanic membrane** which then transmits vibrations into…
  - Vital auditory organs including the three little bones of the **ossicles**
    - The **malleus (hammer)**, the **incus (anvil)**, and the **stapes (stirrup)**

? Review from above: what were the essential sensory cells of the visual system?

- (●) A) Photoreceptors
- ( ) B) Retina
- ( ) C) Pupil
- ( ) D) The optic nerve

Explanation:

- Past the initial pinna, tympanic membrane, and ossicles, there is the **cochlea**, which contains hair structures that function as the ________________ sensory cells of the auditory system

- The cochlea as a whole reminds me of a snail with the head of a cuttlefish
- The stapes (last part of the __________) presses into the **oval window**, which in turn moves fluid around the **hair cells** embedded in the **basilar membrane** tissue in the __________
- The mechanical movement of the hair causes the activation of neurons, __________ their electrical impulses as a signal towards the brain

Discuss: What do we call the state when a neuron is 'ready' to fire? What characteristics are associated with that state?

? Recall from biopsychology, which area of the cortex is most likely to recieve and process information about incoming auditory stimuli (hint, the one on the side of our heads!)

- (●) A) Occipital
- ( ) B) Parietal
- ( ) C) Frontal
- ( ) D) Temporal

Explanation:

5.3 Pitch Perception

- Earlier we discussed how different __________ and wavelengths are connected to the perception of high or low pitch. There are several theories that go further to explain this:
  - The **temporal theory** suggests that the speed of the firing of the __________ cells determines the perceived frequency, but neurons can only fire so fast

Discuss: What is the period after an action potential in which a neuron cannot fire again for a short period of time?

- **Place theory** instead suggests that hair cells on different parts of the ______ membrane are sensitive to different pitch sounds, an activation of a certain part of that membrane are responsible for perception of pitch
  - Under this theory, low-pitch sensors are towards the tip of the membrane, and high-pitch sensors are towards the ______
- Both theories has decent evidence, suggesting they both likely hold some amount of truth

5.4 Sound Localization

? What did we call a visual cue that involves only one eye?

- (●) A) Binocular
- ( ) B) Monocular
- ( ) C) Triocular
- ( ) D) Basal

Explanation:

- In the auditory system, we have **monaural (one-ear)** and **binaural (two-ear)** cues that aid us primarily in ______ sound
- The shape of our ______, or external ear, plays a large role in helping us detect sound!
  - Monocular cues help us detect locate sounds on our ______ axis: directly front, behind, above, and below us
  - Binaural cues consist of **interaural level difference**, or ______ intensity of a sound to our ears, and **interaural timing difference**, when a sound wave reaches each of our ears; these cues help us on the horizontal

access

> **!** Important
>
> So our different cues for vision help us with depth, and our cues for audio help us with position.

5.5 The Vestibular Sense, Proprioception, and Kinesthesia

- **Vestibular sense** is how we maintain ____________ and body posture, and is, surprisingly housed mostly in the inner ear

> **?** Previously, we identified part of the hindbrain that has a lot to do with balance - which structure was that?
>
> - (●) A) Suprachiasmatic Nucleus
> - ( ) B) Broca's Area
> - ( ) C) Thalamus
> - ( ) D) Cerebellum
>
> Explanation:

- Earlier we talk about the inner organ of the ear, but ignore several structures like the **utricle**, **saccule**, and the **semicircular canals**
  - These also have hair cells, just like the ____________ membrane, but instead of being used for our awareness of sound, they are used to help us balance ourselves.
- These structures also contribute to **proprioception (perception of body position)** and **kinesthesia (perception of the body's movement through space)** through those same hair like molecules

6 The Other Senses

6.1 Introduction

> **Important**
>
> Vision and hearing are just two of our complex senses, we have many more! We'll pay less attention to them here, but you can dig in more in your book if you are interested!

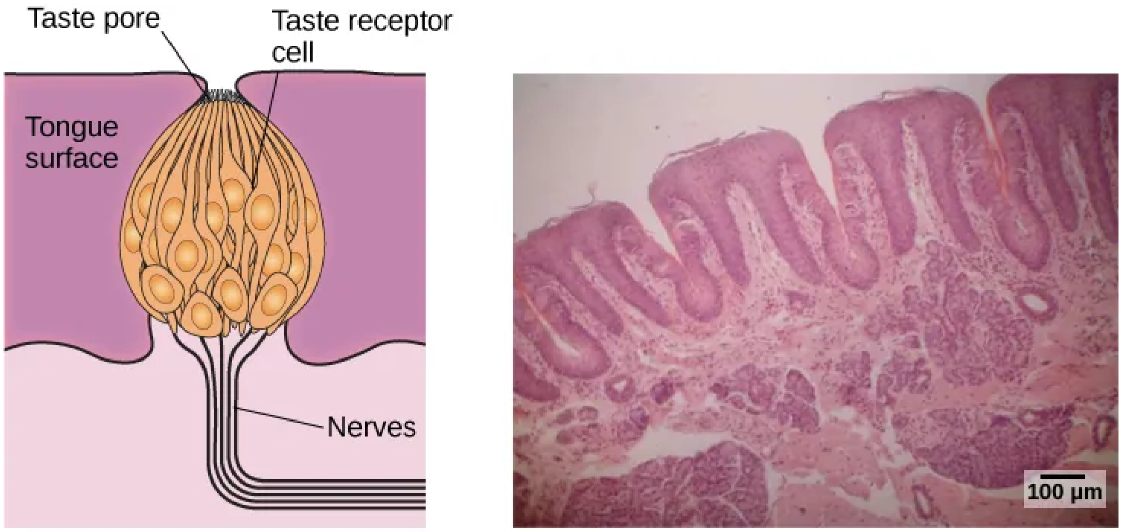
- While vision and audition both work on sensing of __________ through different mechanisms, our senses have different processes to sense

6.2 The Chemical Senses

- Our __________ of taste and smell both work by chemical receptors that are detected in food/drink and the air
  - They are also very interconnected

6.2.1 Taste (Gustation)

- We have six taste groups:
  - __________
  - __________
  - __________
  - __________
  - __________ (Rough translation is "yummy", connected to MSG)
  - Sense of __________ in food
- This happens in the mouth, where **taste buds** on our tongue recieve information as our saliva breaks down food
  - Taste buds have a surprisingly short life span, which is why burning your tongue doesn't have a permanent effect!

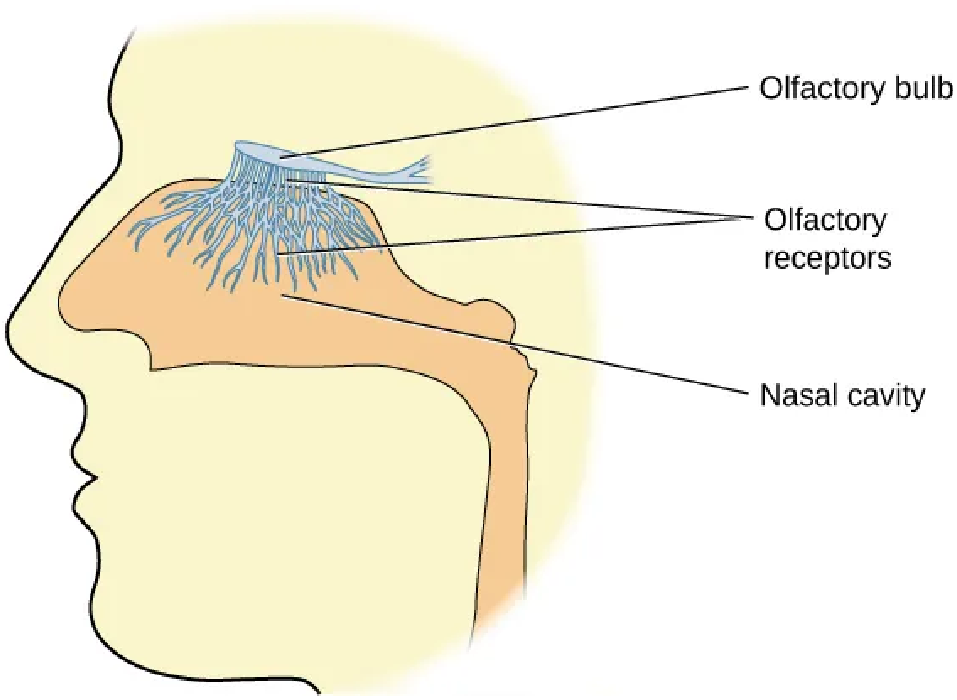


(a) (b)

- Taste, after being routed through the ______________, ends up being processed on the gustatory cortex, near the intersection of the frontal and temporal lobes

6.2.2 Smell (Olfaction)

- Smell works on a similar principle to taste, where chemical odor molecules bind to sites on olfactory ______________, which in turn, sends signals to the **olfactory bulb**

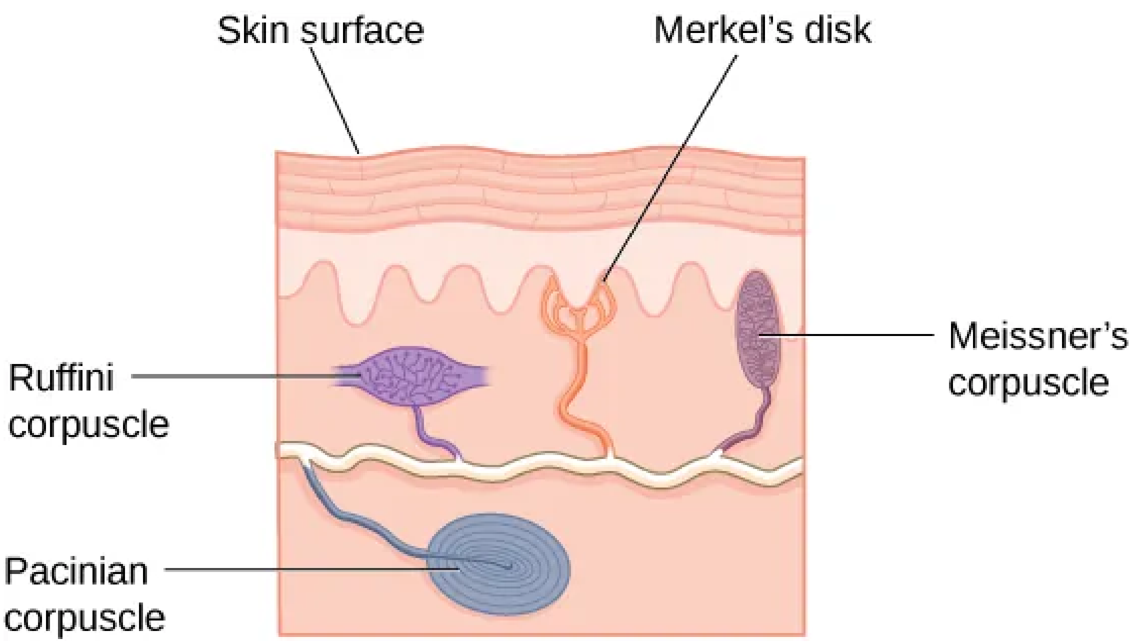


- Smell is processed on the **primary olfactory cortex**, near the ________________ cortex for taste processing

> **Important**
>
> This is one of many reasons that our sense of taste and smell are very related!

6.3 Touch, Thermoception, and Nocicieption

- Our ________________ has many receptors to send signals related to different stimuli
  - **Meissner's corpuscles** → pressure and ________________ frequency vibration
  - **Pacinian corpuscles** → ________________ frequency vibration
  - **Merkel's disks** → ________________ pressure
  - **Ruffini corpuscles** → ________________ sensation



- Free nerve endings (those without specialized names) handle **thermoception (temperature)** and **nociception (harm/pain)**

? All of these skin and touch based signals are processed on the somatosensory cortex of which lobe?

- (•) A) Frontal
- ( ) B) Temporal
- ( ) C) Occipital
- ( ) D) Parietal

Explanation:

6.3.1 Pain Perception

Important

Pain is not universally a 'bad' thing, it keeps us safe and helps us avoid further injury or damage to our body!

- In general, there are two classification of pain:
  - **Inflammatory pain** comes from some type of __________ damage
  - **Neuropathic pain** comes from damage __________ to the neurons of the CNS and/or PNS
- __________ for pain can involve many techniques, including:
  - Talk therapy (pain psychology)
  - OTC or doctor-prescribed __________
  - __________, device implants (deep-brain-stimulation)
  - __________ methods: acupuncture, chiropractics, herbal remedies, etc.

Important

Treatment for pain can vary quite a bit across cultures, where some 'Western Medicine' places emphasis on use of medication, and others place value more in other methods of healing.

- **Congenital analgesia (insensitivity to pain)** is a rare [

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}}"} condition that results in a reduced (or completely diminished) sense of pain.

- This is a very dangerous condition, as the body's "warning system" is effectively not working properly

7 Conclusion

7.1 Recap

- Our senses serve as the basic way that we recieve information about our world, and our sensory organs have complex structures and processes to aid in converting those stimuli into neuronal communications
- Biopsychology remains important, as our modern understanding of sensation is closely tied to brain structure and processing!
- Our perception is our integration and interpretation of sensory information, helpful in not just receiving, but understanding the external stimuli we encounter. Perception does not always contain all of the information transmitted by our senses, nor would we want it to!

7.2 Lecture Check-in

- Make sure to complete and submit the lecture check-in